

As an AI Solution Architect, I have evaluated your Banking Management System. While core banking functions (ledgering, auth, transaction processing) are deterministic and should remain code-based, **AI is highly recommended** to provide the competitive edge required in modern fintech.

1. AI Overview

AI is necessary for **FR-06 (Fraud Detection)** and **FR-07 (Financial Insights)**. Using traditional rules-based systems is insufficient for modern fraud (which is adaptive) and for providing personalized customer experiences.

2. Models

- **Fraud Detection:** *XGBoost* or *Random Forest* (Tabular data classification) for high-speed transaction scoring.
- **Financial Insights (LLM):** *GPT-4o* or *Claude 3.5 Sonnet* via API for natural language budgeting advice.
- **Embedding Model:** *text-embedding-3-small* (OpenAI) to vectorize user spending categories for semantic search.

3. Prompt Engineering

- **System Prompt:** "You are a senior financial advisor. Analyze the user's spending data for the past 30 days. Categorize expenses and provide actionable, empathetic advice for saving. Never provide investment advice; stick to budgeting. Use the user's preferred language."
- **Constraint:** Include a "Disclaimer" prefix in every output stating that insights are for informational purposes only.

4. RAG Design

- **Knowledge Base:** Ingest regulatory documents, bank policy manuals, and public financial literacy articles into a RAG pipeline.
- **Benefit:** When a customer asks, "What are the rules for a personal loan?", the AI retrieves the exact current policy, ensuring the advice is compliant and contextually accurate.

5. Memory

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- **Short-term:** Maintain conversation history for the current session to ensure context continuity.
 - **Long-term (Vectorized):** Store summarized user spending patterns and past interactions to provide continuity in advice (e.g., "Last month you spent \$X on dining, let's try to reduce that by 10%").

6. Agents

- **Support Agent:** An agent that handles common FAQs and redirects to human support if the sentiment score drops below a threshold.
- **Compliance Agent:** An automated auditor that reviews logs periodically to flag anomalies that might violate regulatory guidelines.

7. Vector Database

- **Choice:** *Pinecone* or *pgvector* (PostgreSQL extension).
- **Usage:** Storing vectorized user financial profiles and RAG chunks. *pgvector* is preferred to keep banking data centralized within the existing relational database ecosystem.

8. Embeddings

- **Categorization:** Convert transaction descriptions (e.g., "Starbucks #123") into semantic vectors to automatically group them into buckets like "Dining" or "Entertainment," improving upon basic keyword matching.

9. LLM Workflow

1. **Ingestion:** Real-time transaction ingestion -> Scrub PII.
2. **Analysis:** Tabular ML model predicts fraud likelihood.
3. **Synthesis:** LLM fetches user history and current context.
4. **Generation:** AI generates a personalized insight.
5. **Human-in-the-loop:** Fraud alerts marked "High" are routed to the Admin Dashboard for manual review before action.

10. AI Risks

- **Hallucinations:** AI might suggest inaccurate loan eligibility. **Mitigation:** Force the LLM to ground answers in the "Loan Policy" document using RAG; reject all math-based calculations from the LLM—perform calculations in the backend code and pass the result to the LLM.

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- **Bias:** Training data could bias loan approval suggestions. **Mitigation:** Use "Fairness Toolkits" (e.g., IBM AI Fairness 360) to audit model output for protected group bias.
 - **Data Leakage:** Financial data leaking into training sets. **Mitigation:** Use Private/Enterprise API instances (e.g., Azure OpenAI) where data is not used to train global models.